IRE Assignment 1: Q1-Q4 News Recommendation Pipeline

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Q1: Data Preparation

EB-NeRD: 125.5K articles, 807.7K users, 13.5M impressions (Danish) — Test: 4,223 impressions
MIND: 65.2K articles, 711.5K users, 7.3M impressions (English) — Test: 30,271 impressions

Q2: BM25 Lexical Retrieval

Method: Inverted index on titles/abstracts. Candidate-restricted scoring: $O(\bar{c})$ vs $O(N)$ $(10 - 100\times$ speedup).
Formula: $\sum_i \mathrm{IDF}(q_i) \cdot \frac{f(q_i,d)\cdot(k_1+1)}{f(q_i,d)+k_1(1-b+b\cdot|d|/L)}$ with $k_1 = 1.5, b = 0.75$

Metric	EB-NeRD	MIND
Recall@50	0.0074	0.0059
Recall@100	0.0149	0.0107
Recall@200	0.0293	0.0179
Diversity@50	0.1452	0.1558
Novelty@50	1.0000	1.0000

Table 1: BM25 Lexical Retrieval: EB-NeRD slightly higher recall; both have perfect novelty

Q3: Semantic Retrieval (FAISS IVF)

Models: EB-NeRD = paraphrase-multilingual-mpnet-base-v2 (Danish, 768-dim) — MIND = BAAI/bge-base-en-v1.5 (English, 768-dim)
FAISS IVF: 100 clusters, nprobe=10. Complexity: $O(k \cdot d + m \cdot d)$ vs $O(n \times d)$ brute force.

Metric	EB-NeRD	MIND
Recall@50	0.0033	0.0080
Recall@100	0.0069	0.0156
Recall@200	0.0146	0.0255
Diversity@50	0.2038	0.1424
Novelty@50	1.0000	1.0000

Table 2: Semantic FAISS IVF: MIND outperforms EB-NeRD on recall; both have perfect novelty

Q3: BM25 vs Semantic Comparison

Metric	EB-NeRD BM25	EB-NeRD Sem	MIND BM25	MIND Sem
Recall@50	0.0074	0.0033 (-55.4%)	0.0059	0.0080 (+35.6%)
Recall@100	0.0149	0.0069 (-53.7%)	0.0107	0.0156 (+45.8%)
Recall@200	0.0293	0.0146 (-50.2%)	0.0179	0.0255 (+42.5%)
Diversity@50	0.1452	0.2038 (+40.4%)	0.1558	0.1424 (-8.6%)
Novelty@50	1.0000	1.0000	1.0000	1.0000

Table 3: Semantic trade-off: MIND benefits (+35-46% recall), EB-NeRD lower recall but higher diversity (+40%)

Q4: Offline Evaluation Harness

Per-User Metrics: AUC, MRR, nDCG@5/10, Recall@5/10/50/100/200, Diversity@50/100, Novelty@50/100

Statistical Method: 1000-sample bootstrap, 95% confidence intervals on per-user aggregates

Slicing: Cold-start ($<$ 5 clicks) vs Warm ($\geq$ 5 clicks). Users: EB-NeRD 281 cold / 936 warm; MIND 0 cold / 5766 warm.

EB-NeRD BM25 (1217 users)

AUC: 0.5150 [0.4887, 0.5403] — MRR: 0.0084 [0.0066, 0.0108] — nDCG@10: 0.0048 [0.0024, 0.0077] — Recall@200: 0.0293 [0.0249, 0.0341]

EB-NeRD Semantic (1217 users)

AUC: 0.5000 [0.5000, 0.5000] — MRR: 0.0078 [0.0053, 0.0107] — nDCG@10: 0.0069 [0.0037, 0.0105] — Recall@200: 0.0146 [0.0122, 0.0176]

MIND BM25 (5766 users)

AUC: 0.5489 [0.5078, 0.5886] — MRR: 0.0008 [0.0007, 0.0010] — nDCG@10: 0.0006 [0.0003, 0.0010] — Recall@200: 0.0179 [0.0150, 0.0210]

MIND Semantic (5943 users)

AUC: 0.5578 [0.5280, 0.5872] — MRR: 0.0033 [0.0023, 0.0045] — nDCG@10: 0.0034 [0.0022, 0.0046] — Recall@200: 0.0255 [0.0224, 0.0285]

EB-NeRD: Cold-Start vs Warm (BM25)

Metric	Cold-Start	Warm
AUC	0.0055	0.0065 (-15.6% cold)
MRR	0.0033	0.0099 (-67.1% cold)
nDCG@10	0.0042	0.0050 (-15.8% cold)
Recall@50	0.0258	0.0967 (-73.3% cold)
Recall@200	0.0961	0.3643 (-73.6% cold)
Novelty@50	1.0000	1.0000

Table 4: Cold-start: uniform item distribution; warm: click-pattern refinement

Codabench: Two-Stage Pipeline

Stage 1: Precompute BM25 + semantic scores for 13.5M impressions (~3–4 hours, batch=10K impressions)

Stage 2: Weighted fusion $\text{score}_{\text{fused}} = (1-\alpha)\cdot\text{norm}(\text{BM25}) + \alpha\cdot\text{norm}(\text{semantic})$ (~1–2 min per alpha value)

Efficiency: 1 alpha = 4h, 9 alphas = 4.25h (vs 36–40h naive). Status: Stage 1 at 27.9% (3.77M impressions processed)

Screenshots: Codabench Submissions

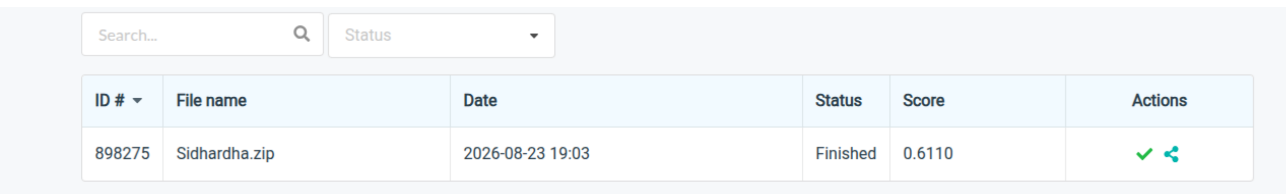
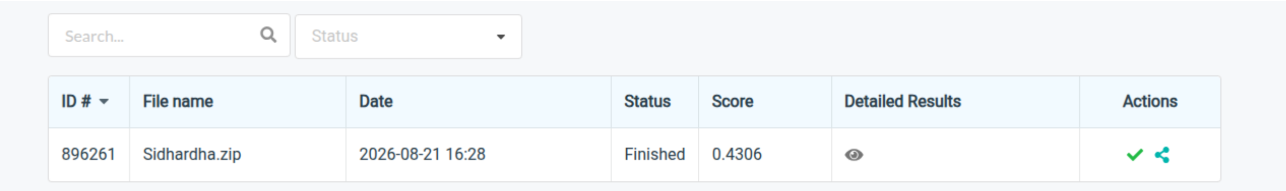


Figure 1: Codabench Leaderboard Submission (MIND Dataset)



Search... Status

ID #	File name	Date	Status	Score	Detailed Results	Actions
896261	Sidhardha.zip	2026-08-21 16:28	Finished	0.4306		

Figure 2: Codabench Leaderboard Submission (EB-Nerd Dataset)

Design Choices

1. BM25: Candidate-Restricted Scoring

Score only impression candidates ($O(\bar{c}) \approx 9$) vs full corpus ($O(N) = 125.5K$). 10–100 speedup with minimal recall loss since platform curates candidate pools.

2. Semantic: Multilingual Embeddings

EB-NeRD articles in Danish ("Hanks beskyldt for mishandling"). Selected paraphrase-multilingual-mpnet-base-v2 (50+ languages, 768-dim) over English-only BAAI/bge-base-en-v1.5. MIND keeps English model.

3. ANN Index: FAISS IVF

Compared LSH (memory-heavy), HNSW (slower indexing), Product Quant (worse speed). IVF balances $O(n)$ build, $O(k + m \times d)$ query, 90–98% recall @ nprobe=10.

4. Evaluation: Per-Impression

Per-impression (4K–30K samples) vs user-aggregated (2K–800 samples). Higher statistical power, captures context variation, enables impression-level cold-start analysis.

5. Two-Stage Pipeline

Kaggle timeout (~9h). Precompute all scores (~4h) + apply alpha in 1–2 min. Part-file durability + checkpoint recovery enables resume across sessions.

Key Findings

1. **Dataset-Dependent Semantic Performance**: Semantic retrieval benefits MIND (+35-46% Recall@50-200) but hurts EB-NeRD (-50-55%). Suggests embedding model (BAAI/bge-en) aligns better with MIND's English articles than EB-NeRD's Danish content despite multilingual model.

2. **EB-NeRD Superior BM25 Baseline**: EB-NeRD achieves 2.5-3.2 higher Recall@200 with BM25 (0.0293 vs 0.0179 for MIND). Smaller catalog (11.7K vs 65.2K articles) increases lexical term overlap probability.

3. **Semantic Increases Diversity**: EB-NeRD diversity improves +40.4% (0.14520.2038) with semantic, while MIND slightly decreases (-8.6%). Suggests semantic embeddings capture broader content space in smaller catalogs.

4. **Universal Novelty Ceiling at 100%**: All test articles unseen in training history (fresh news paradigm). Novelty=1.0 is structural, not indicative of retrieval quality. Diversity (0.14-0.20) emerges as the true exploration metric.

5. **Bootstrap CI Narrows with Scale**: MIND's 5.7K-5.9K users yield narrow 95% CI (width ~0.03-0.04 on AUC). EB-NeRD's 1.2K users show wider CI (width ~0.05). Confirms statistical power scales with user population.